L-9

Expectation $\quad X$

$$E(X) = \sum_{\substack{x:\\ P(X=x)>0}} x\, P(X=x)$$

$X \sim \text{Bern}(p) \quad ; \; E(X) = p$

$X \sim \text{Bin}(n,p) \quad ; \; E(X) = np$

Linearity of Expectation:

(i) $E(X+Y) = E(X) + E(Y) \longrightarrow$

(ii) $E(cX) = c\,E(X) \quad ; \; c > 0$

$E(X+Y)$

$$\sum_{\substack{t:\\ P(X+Y=t)>0}} t\, P(X+Y=t)$$

$$P(X+Y=t) = \sum_{x\,:\,P(X=x)>0} P(X+Y=t \mid X=x)\, P(X=x)$$

$$\cancel{Y} \; \overset{?}{=} \sum_{x} \underbrace{P(Y = t-x \mid X=x)}\, P(X=x)$$

$$E(X) = \sum x\, P(X=x)$$

$\{\{HH\}, \{HT\}, \{TH\}, \{TT\}\}$

$\{HH\}$	$\{HT\}$, $\{TH\}$	$\{TT\}$
$X=0$	$X=1$	$X=2$

$$E(X) = \sum_{s \text{ event}} X(s) P(\{s\}) = \sum x P(X=x)$$

$$E(X+Y) = \sum (X+Y)(s) P(\{s\})$$
$$= \sum X(s) P(\{s\}) + \sum Y(s) P(\{s\})$$
$$= E(X) + E(Y)$$

Negative Binomial

$X \sim NBin(r, p)$

r

$r = 5$ success

Geometric Distribution

X = # of trials until you get 1st success

Y = # of failures before the 1st successful trial

$Y = X - 1$

$$P(Y=k) = ? \quad P(X = k+1)$$
$$= (1-p)^k p = q^k p$$

$$E(Y) = E(X-1)$$
$$= E(X) - E(1) = E(X) - 1$$
$$= \frac{1}{p} - 1$$
$$= \frac{q}{p}$$

($E(c) = c$ for any constant c)

Negative Binomial

$X \sim NegBin(r, p)$

X = # of trials needed to obtain the rth success

$$\boxed{P(X=k) = {}^{k-1}C_{r-1}\, p^r (1-p)^{k-r} \quad k \geq r}$$

$k = r, r+1, \ldots,$

Ex.

Y = # failure before the rth success

Find what is PMF of Y.

$$E(X) = \sum_{k=r}^{\infty} k\, P(X=k)$$

$X = X_1 + X_2 \ldots + X_r$

X_j = # failure between $j-1$ & jth success (follows geometric distribution

$= \frac{rq}{p}$

Poisson distribution $X \sim Pois(\lambda)$, $\lambda \geq 0$

PMF of $X = P(X=k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$

$k \geq 0$

λ = "rate" parameter.

Valid PMF = $\sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} e^{\lambda}$

$= 1$

$$E(X) = \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}$$

$$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1} \cdot \lambda}{(k-1)!}$$

$$= \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda$$

Situations where applicable

large no. of trials each with small probability of success

1. # emails in an hour
2. # earthquakes in a year in some region

Poisson distribution is a suitable model for events which

(i) occur randomly in space/time
(ii) cannot occur simultaneously
(iii) occur independently
(iv) large no. of trials & probability of success is very small.

a) # of cars per minute passing under a road bridge betⁿ 10 am - 11 am, when the traffic is flowing freely.

b) # of cars entering a mall's parking on a busy Sunday

c) # of blood cells per ml in a dilute solution of blood which has been left standing for 24 hrs.

d) # of blood ~~cls~~ cells per ml in a

well-shaken dilute soln of blood.

L-10

Recall, $Y = Y_1 + Y_2 + \cdots + Y_r$

Negative Binomial : $X \sim NBin(r, p)$

# of trials / no. of failures to get r successes

$$P(X = k) = {}^{k-1}C_{r-1}\, p^r (1-p)^{k-r}$$

$$P(Y = k) = {}^{k+r-1}C_{r-1}\, p^r (1-p)^k$$

$$E[Y] = \frac{rq}{p}$$

To derive this, FFFSFFS

$Y = Y_1 + Y_2 + \cdots + Y_r$;

Y_1 = # failures before 1st success

Y_2 = # failures btw 1st & 2nd success

$\vdots$

Y_r = # failures " r-1 th & r th success

Each $Y_i \sim Geom(p)$

$$E[Y_i] = \frac{q}{p}$$

$X(\omega) = c$

$E[c] = c$ $E[c]$ $E[X]$

$X(\omega) = c$

$E(c) = E[X]$

$= \sum x P(X = x)$

$E[Y] = \frac{rq}{p}$

$X = Y + r$

$E[X] = E[\underbrace{Y + r}]$

$= E[Y] + r$

$P(X=x) > 0$

$= c \sum_{x:\, P(X=x)>0} P(X=c)$

$= c$

$X = Y + r \Rightarrow E[X] = \underset{= \frac{rq}{p}}{E[Y]} + r = r\left(\frac{q+p}{p}\right) = \frac{r}{p}$

Poisson distribution : $X \sim Pois(\lambda)$ $\quad \lambda > 0$

PMF of X = $P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$; $k \geq 0$

$E[X] = \lambda$ $\qquad$ Probability of k no. of events happening

<u>Models</u> :

times a specific event occurs within a fixed interval of time/space.

In other distributions : fixed number of trials

In Poisson : Dont know number of trials but know

λ = avg. rate at which events happen over a specific interval

To use the model, we should have

① Independent events : 1 event happening does not change the probability of another happening.

(1 car crash)

(2) Events do not happen simultaneously

(3) Constant rate : The average rate λ stays the same throughout the interval.

(4) Events are random & rare.

Catastrophes, typos, accidents,

no. of wrong telephone nos. that are dialed in a day, . . .

$X \sim Bin(n, p)$

PMF of $X = P(X = k)$

$$= {}^{n}C_{k} \cdot p^{k} (1-p)^{n-k}$$

$n \to \infty, \quad p \to 0$

$\lambda = np$ ↳ constant

$$= \frac{n!}{(n-k)!\,k!} \cdot \left(\frac{\lambda}{n}\right)^{k} \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

$$= \frac{n(n-1)\cdots\cdots(n-k+1)}{n^{k}} \frac{\lambda^{k}}{k!} \cdot \left(1-\frac{\lambda}{n}\right)^{n} \left(1-\frac{\lambda}{n}\right)^{-k}$$

$\lim_{n \to \infty}$

1

$\left(1-\frac{\lambda}{n}\right)^{-k}$

$\overset{n\to\infty}{\Rightarrow} \quad 1 \cdot \frac{\lambda^{k}}{k!} \cdot 1 \cdot \left(1-\frac{\lambda}{n}\right)^{n} \longrightarrow \infty \quad n \to \infty$

$$e^{x} = \lim_{n\to\infty} \left(1+\frac{x}{n}\right)^{n}$$

$$\approx 1 \cdot \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda}$$

HW

$$\approx \frac{e^{-\lambda} \lambda^k}{k!}$$

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

For large n, & p small

$$P(X = k) \approx \frac{e^{-\lambda} \lambda^k}{k!}$$

$$\lambda = np$$

Ex. avg. rate of accidents per week = 3

probability that there is atleast 1 accident this week.

$\lambda = 3$

X = # of accidents this week

$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - \frac{e^{-3} \cdot 3^0}{0!}$$

$$= 1 - e^{-3}$$

$$\approx 0.95$$

Ex. Suppose the probability that an item produced will be defective = 0.1

Find the probability that a sample of 10 items will contain at most 1 defective item.

Soln: $P(X=0)+P(X=1)$

$${}^{10}C_0 \cdot (0.1)^0 (0.9)^{10} + {}^{10}C_1 (0.1)^1 (0.9)^9$$

$$\simeq 0.7361$$

$\lambda = np = \cancel{10 \cdot 0.1}$

$= 10 \cdot 0.1 = 1$

$$\frac{e^{-1} \cdot 1^0}{0!} + \frac{e^{-1} \cdot 1^1}{1!} \simeq 0.7358$$

Ex. $n \to$ distinct figure

$T =$ # of toys to collect so that we have the entire collection of figures

$T = T_1 + T_2 + \dots + T_n$

$T_i \sim \text{Geom}(p_i)$

Ex. p_i

T_2 $\quad p_2 = \frac{n-1}{}$

$T_3 \qquad p_3 = \frac{n-2}{n}$

$$p_i = \frac{n-(i-1)}{n} = \frac{n-i+1}{n}$$ Complete this!

$E[T_i] =$

$$E[T] = \sum_{i=1}^{n} E[T_i]$$

$\boxed{L-11}$

$X \to$ r.v

$E(X) = \sum x P(X=x)$

$E(X^2)$

$X \to \{0, 1, 2, \ldots\}$ $\quad p_0, p_1, p_2 \ldots \quad P(X=n) = p_n$

$X^2 \to \{0, 1, 4, 9, \ldots\}$

$\quad p_0, p_1, p_2 \ldots$

$P(X^2 = k)$

$$E(X) = \sum_{n=0}^{\infty} n p_n$$

$$E(X^2) = \sum ?$$

$$\sum n^2 p_n$$

Law of the unconscious statistician (LOTUS)

$$E(g(X)) = \sum g(x) P(X=x)$$

$$E(g(X)) = \sum g(x) P(X=x)$$

$$E(X+Y) = E(X) + E(Y)$$

$$E(X) = \sum x P(X=x) = \sum_{s \in S} X(\{s\}) P(\{s\})$$

$(\{HT\}, \{HH\}, \{TT\}, \{TH\})$

group

ungroup

$$E(g(X)) = \sum_{s \in} g(X(\{s\})) P(\{s\})$$

$$= \sum_x \sum_{s : X(\{s\}) = x} g(x) P(\{s\})$$

$$= \sum_x g(x) \left(\sum_{s : X(s) = x} P(\{s\}) \right)$$

$$\boxed{E(g(X)) = \sum_x g(x) P(X=x)}$$

$E(X)$

$$E(X - E(X)) = 0$$

$$E(X-c) = E(X) - E(c)$$
$$= E(X) - c$$
$$= E(X) - E(X) = 0$$

$$\mathrm{Var}(X) = E\left((X - E(X))^2\right)$$

$$E(cX) = cE(X)$$

$$= E\left(X^2 + (E(X))^2 - 2XE(X)\right)$$
$$= E(X^2) + (E(X))^2 - 2E(X)E(X)$$
$$= E(X^2) + (E(X))^2 - 2(E(X))^2$$
$$= E(X^2) - (E(X))^2 \geq 0$$

≥ 0

$$E(g(X)) = \sum g(x)\, P(X=x)$$
$$E(X^2) = \sum x^2\, P(X=x)$$

$\boxed{L-12}$

① (a) A batch of 15 blood samples contains 6 infected samples and 9 healthy samples. b
A lab technician tests samples one by one, without replacement, and stops as soon as the 2nd infected sample is detected.
Find the probability that the number of samples inspected until the 2nd infected sample is found is 5.

R.V.

Y = # of samples drawn to get 2 infected samples

Suppose we did k draws to get 2 infected samples

Previous $k-1$ draws — 1 infected draw → $^{6}C_{1}$

— $k-2$ healthy → $^{9}C_{k-2}$

Total outcomes $^{15}C_{k-1}$

k th draw → 1 infected → $\frac{5}{15-(k-1)}$

$$P(X=k) = \frac{^{6}C_{1} \cdot {}^{9}C_{k-2}}{^{15}C_{k-1}} \quad \frac{5}{15-(k-1)}$$

$P(X=5)$

Generally,

$k-1$ → $r-1$ infected

→ $(k-1)-(r-1)$, $k-r$ heath

r th

k sample

(b) Derive a general formula for the PMF when you have 'g' infected samples and 'h' healthy samples, & the technician stops until he finds 'r' infected samples.

$$P(X=k) = \frac{^{g}C_{r-1} \cdot {}^{h}C_{k-r}}{^{g+h}C_{k-1}} \quad \frac{g-(r-1)}{(g+h)-(k-1)}$$

$k = r, r+1, \ldots, h+r$

Functions of random variables

$X : S \to \mathbb{R}$ sample space

$X(s) \in \mathbb{R}$, $s \in S$ outcome

Ex. $S = \{HT, TH, HH, TT\}$ outcome $\{HT, TH\}$

$X = \#$ of heads in an outcome

S sample space

Probability function $P : \mathcal{P}(S) \to [0,1]$

Events : elements of the power set of S

(i) $P(\emptyset) = 0$, $P(S) = 1$

(ii) $A_1, A_2, \ldots,$ mutually disjoint

$$P\left(\bigcup A_i\right) = \sum_{i \geq 1} P(A_i)$$

Outcomes : elements of S

Random variable $X : S \longrightarrow \mathbb{R}$

$(X(s) \in \mathbb{R}$

Ex. $S = \{HT, TH, TT, HH\}$

Expeiment : 2 Coin tosses

Power set of $S = \{\emptyset, \{HT\}, \{TH\}, \{TT\}, \{HH\},$
$\{HT, TH\}, \{TT, HH\}, \ldots$

$\{HT, TH, TT\}, \ldots, S\}$

What are the possible outcomes for the following event?

Q.1 There should be atleast 1 head?

Q.2 Same face should not appear on both coins

Q.3 Both land on tail

$X: S \to \mathbb{R}$

$g(x) = x^2$ $\quad g: \mathbb{R} \to \mathbb{R}$

$g(X): S \to \mathbb{R}$

$g(X) = X^2$

Functions of random variable

g

$X: S \to \mathbb{R}$, $\quad g: \mathbb{R} \to \mathbb{R}$

$g(X): S \to \mathbb{R}$

Then,

X

$g(X)$ is a random variable $g(X)(s) \in \mathbb{R}$

$g(x) = x^2$

$g(X) = X^2$

$g(X): S \to \mathbb{R}$

$s \in S$

$\cancel{X}(s) \in \mathbb{R}$

$g(X)(s) := g(X(s))$

1. How to find PMF of $g(X)$
2. How to find $E(g(X))$

1. PMF of $Y = g(X)$:

$$P(Y = y) = P(g(X) = y)$$

Range of $g(X)$?

$y \in$ Range of $g(X)$ iff

$y = g(x)$ for some $x \in$ range of X.

Let $y \in$ Range of $Y = g(X)$.

$$P(Y=y) = P(g(X)=y)$$
$$= P(\{s \in S : g(X)(s) = y\})$$
$$= P(\{s \in S : g(X(s)) = y\})$$
$$= P\left(\bigcup_{x : g(x)=y} \{s \in S : X(s) = x\}\right)$$

Note that each set above is disjoint.

If $x_1 \neq x_2$

$$\{s \in S : X(s) = x_1\} \cap \{s \in S : X(s) = x_2\} = \emptyset$$

$$= \sum_{x : g(x)=y} P(\{s \in S : X(s) = x\})$$

$$\boxed{P(g(X)=y) = \sum_{x : g(x)=y} P(X=x)}$$

Ex. A particle moves 'n' steps on a number line. The particle starts at 0, and at each step it moves 1 unit to the right or to the left, with equal probabilities. Assume all steps are independent.

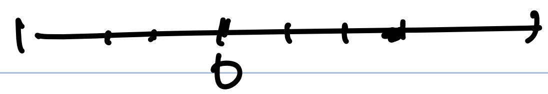
Let D = distance of the particle from the origin after 'n' steps.

Sol^n: Let X_i be the ith step, where

$$P(X_i = 1) = 0.5 \quad (\text{ith step is on right})$$
$$P(X_i = -1) = 0.5 \quad (\text{ith step is on left})$$

The position of the particle after 'n' steps / displacement

$$X = X_1 + X_2 + \cdots + X_n$$



Let Y = # of steps to the right

$n - Y$ = # " " " " left

Displacement $= X = Y - (n - Y) = 2Y - n$

Range of $Y = \{0, 1, 2, \ldots, n\}$

Distribution of Y = Binomial$(n, \frac{1}{2})$

$$P(Y = k) = {}^nC_k \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^{n-k}$$

$$\Rightarrow P(X = k) = P(2Y - n = k) = P\left(Y = \frac{n+k}{2}\right)$$

D = Distance $= |X| = |2Y - n|$

Find PMF of $D = |X|$

$$\boxed{\begin{aligned} P(|X| = k) &= \sum_{x : |x| = k} P(X = x) = P(X = k) + P(X = -k) \\ &= P\left(Y = \frac{n+k}{2}\right) + P\left(Y = \frac{n-k}{2}\right) \end{aligned}}$$

2. How to find $E[g(X)]$?

let $Y = g(X)$

$$E[g(X)] = \sum_{y:\, y \in \text{Range}(g(X))} y \, P(Y = y)$$

$$= \sum_{y} y \sum_{\substack{x: \\ g(x) = y}} P(X = x)$$

$$= y_1\big(P(X = x_1) + P(X = x_2)\big) + y_2\big(P(X = x_3) + P(X = x_4)\big)$$

$g(x_1) = y_1 = g(x_2)$ $\quad$ $g(x_3) = g(x_4) = y_2$

$$= \sum_{x:\, P(X = x) > 0} g(x)\, P(X = x)$$

Variance of a r.v. X

$$Var(X) = E[(X - E[X])^2]$$

On expanding

$$Var(X) = E(X^2) - (E(X))^2$$

Properties of Expectation & variance

1. $E(X + Y) = E(X) + E(Y)$
2. $E(aX + b) = aE(X) + E(b)$
$= aE(X) + b$

3. $E[g(X)] = \sum_{x} g(x) P(X = x)$

Variance

1. $Var(X + c) = Var(X)$

$$E\left((X+c) - E(X+c) \right)$$
$$E\left(X - E(X) \right) = Var(X)$$

2. $Var(cX) = c^2 Var(X)$

$$E\left[(cX - E[cX])^2 \right] = E\left[c^2 (X - E[X])^2 \right]$$
$$= c^2 Var(X)$$

Prob: Are there discrete random variable X & Y such that $E(X) > 10^2 E(Y)$ but $P(Y > X) \geq 0.99$